Yoav Gur-Arieh

PhD Student researching interpretability in LLMs, with 11 years software engineering experience

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Education

Tel-Aviv University — *Dec 2025 to present*
PhD in Computer Science
- NLP lab, researching interpretability in LLMs under advisor Mor Geva.
- Working on understanding and controlling reasoning processes in LLMs.

Tel-Aviv University — *Oct 2024 to Dec 2025*
MS in Computer Science — summa cum laude
- NLP lab, researching interpretability in LLMs under advisor Mor Geva.
- Done as part of the direct-to-masters interdisciplinary program.
- Average: 98

Tel-Aviv University — *Oct 2022 to Jun 2026*
Interdisciplinary Studies
- Studying in the Adi Lautman Interdisciplinary Program for Outstanding Students.
- Took courses in computer science, biology, neuroscience, physics and history.
- Average: 94

Experience

Graduate Researcher (*Advisor: Mor Geva*) — *Sep 2024 to present*
Tel-Aviv University
- Published research on improvements to large-scale automated **feature interpretability pipelines**, and **machine unlearning**.
- Working on understanding and controlling **reasoning** in LLMs.

AI Researcher — *Feb 2026 to present*
AAI
- Working on designing expert AI systems.

Research Fellow — *Jun 2026 to Aug 2026*
Cambridge Boston Alignment Initiative (CBAI)

Research Intern (*Supervisor: Atticus Geiger*) — *Jun to Sep 2025*
$Pr(Ai)^2R$ Group

Senior Software Engineer — *Mar 2021 to Feb 2026*
Laminar Security / Rubrik
- Joined the startup at its inception, spearheading the development of front-end, back-end, and agent-based systems from scratch, including a document sensitivity classification engine.
- Led technical research and implemented solutions for extracting decrypted data from encrypted cloud traffic.

Senior Developer & Researcher — *Mar 2016 to Mar 2021*
Stealth
- Carried out months-long solo research projects into esoteric and opaque technologies that have little or no publicly available documentation, culminating in the development of technical solutions.
- Led through to completion highly complex, high-risk projects involving multiple teams.

Software Developer — *Aug 2015 to Mar 2016*
Checkpoint Software Technologies
- Contributed to the development of a log aggregator and analyzer product tailored for large enterprises.

Publications

Faithfulness Metrics Don't Measure Faithfulness: A Meta-Evaluation with Ground Truth *May 2026*

Yoav Gur-Arieh, Ana Marasović, Mor Geva

Submitted to NeurIPS 2026
Oral presentation at NEMI 2026

Disentangling MLP Neuron Weights in Vocabulary Space *April 2026*

Asaf Avrahamy, **Yoav Gur-Arieh**, Mor Geva

Accepted to COLM

Mixing Mechanisms: How Language Models Retrieve Bound Entities In-Context *October 2025*

Yoav Gur-Arieh, Mor Geva, Atticus Geiger

Accepted to ICLR 2026

LMEnt: A Suite for Analyzing Knowledge in Language Models from Pre-training Data to Representations *September 2025*

Daniela Gottesman, Alon Gilae-Dotan, Ido Cohen, **Yoav Gur-Arieh**, Marius Mosbach, Ori Yoran, Mor Geva

Accepted to TACL

Precise In-Parameter Concept Erasure in Large Language Models *May 2025*

Yoav Gur-Arieh, Clara Suslik, Yihuai Hong, Fazl Barez, Mor Geva

Accepted to EMNLP Main 2025

Enhancing Automated Interpretability with Output-Centric Feature Descriptions *Jan 2025*

Yoav Gur-Arieh, Roy Mayan, Chen Agassy, Atticus Geiger, Mor Geva

Accepted to ACL Main 2025

Invited Talks

Technion, NLP Group – *Evaluating Chain-of-Thought Faithfulness in LLMs* *Aug 2026*

Harvard University, Kempner Institute PRISM Group – *Evaluating Chain-of-Thought Faithfulness in LLMs* *Aug 2026*

Northeastern University, Bau Lab – *Evaluating Chain-of-Thought Faithfulness in LLMs* *Jul 2026*

Hebrew University, NLP Group – *Mixing Mechanisms: How LMs Use Pointers to Bind and Retrieve Entities* *Jan 2026*

Max Planck Institute for Security and Privacy – *From Description to Erasure: Feature-Based Control of LLMs* *Sep 2025*

Projects

SAE Knowledge Erasure Project *Code*

- Leveraged output-centric feature descriptions to identify MLP SAE features associated with specific concepts in the Gemma-2 2B model. Ablated these features to effectively erase the corresponding concepts, demonstrating targeted knowledge manipulation - explained in detail here.

Avian Neuronal Response Project *Code*

- Analyzed avian neuronal activity in response to varied bird calls (authentic and artificial), and used classical machine learning techniques to classify stimuli.

Parkalot - Parking Finder App *Link*

- Developed an app that displays parking lots with free spots around the user.

Hoppa - Platform Jumper Android Game *Link*

- Developed an android platform jumper game using Unity in C#, from conception to deployment.

Technologies

Languages: Python, C, Golang, C#, Java, JavaScript

Technologies: PyTorch, HF Transformers, TransformerLens, SAELens, eBPF, Linux Internals, Cybersecurity

Language Proficiency

English, Hebrew: Native